Amplifiers

SAGI Electronics Lecture 6

Amplifier Fundamentals

- An amplifier is a 2-port device that uses an external power supply to increase an electrical signal.

DC Power Supply

Weak Input Signal

Boosted Output Signal

Amplifier Fundamentals

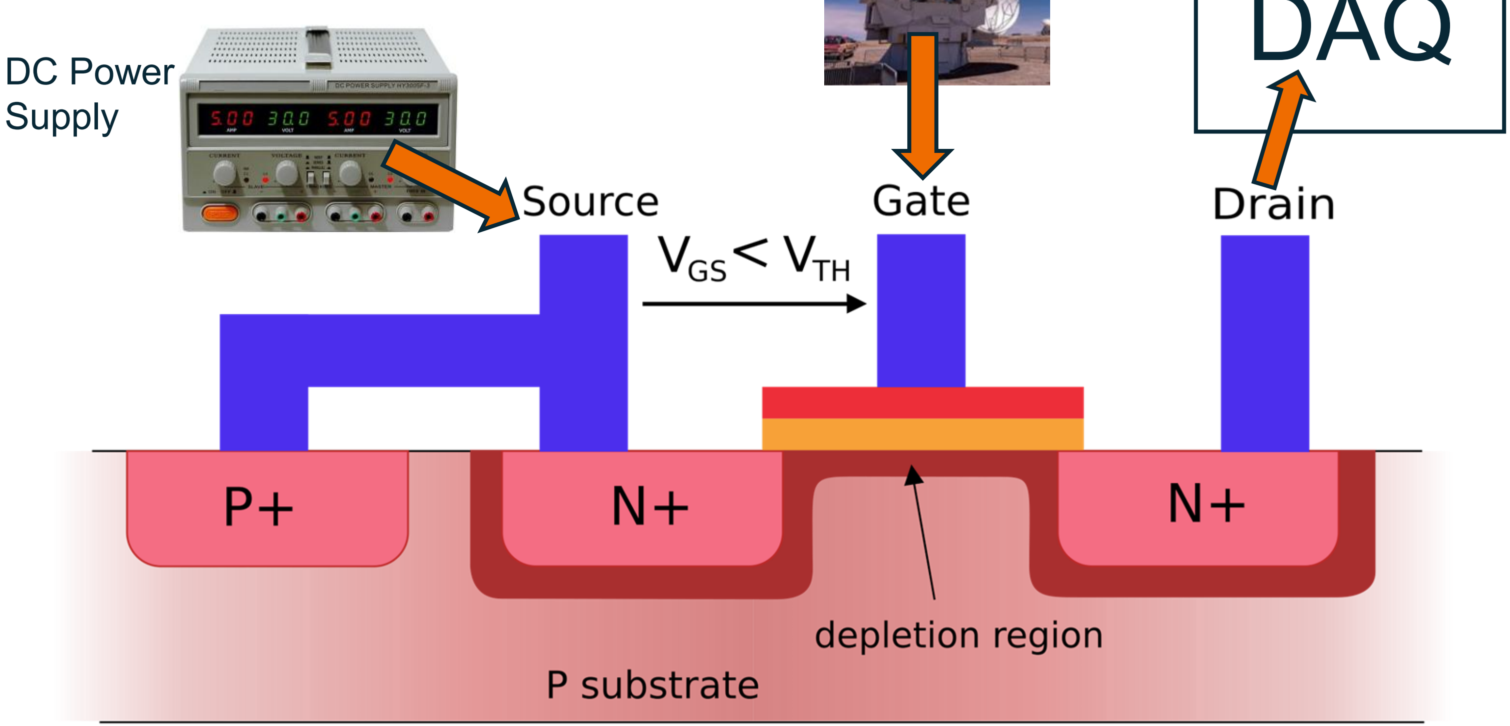
What would we consider as we’re picking an amplifier?

Amplifier Fundamentals

What would we consider as we're picking an amplifier?

- Gain
- Noise
- Dynamic range
- Input/output impedance
- Bandwidth
- Phase shift
- Power dissipation
- Cost
- Temperature stability
- ...

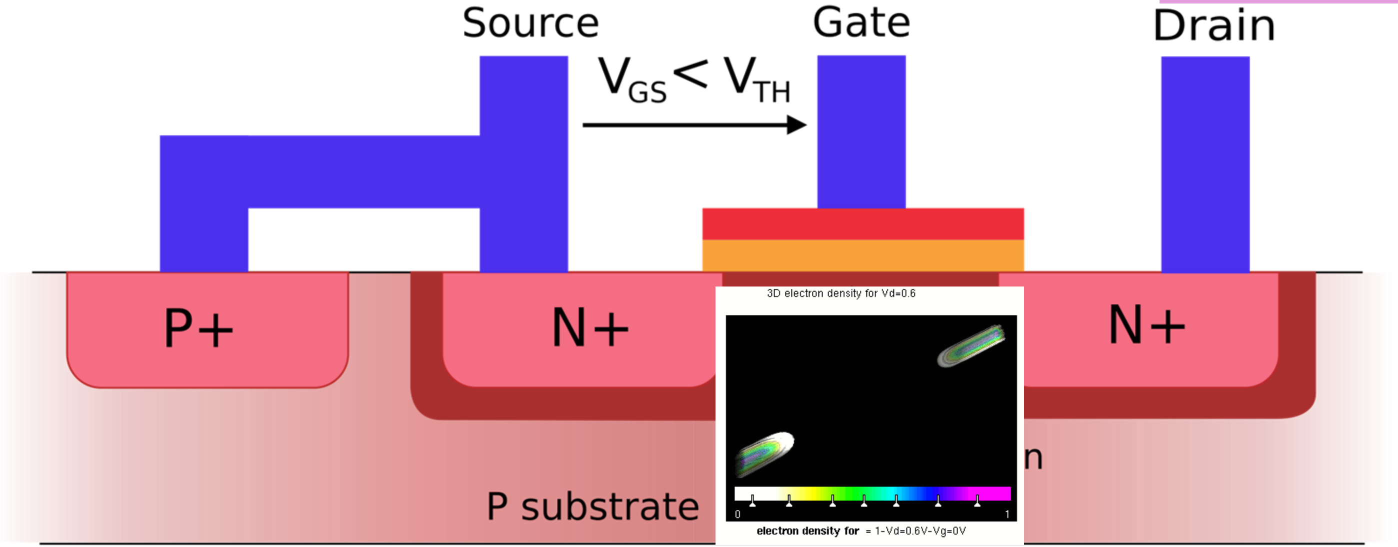
Amplifier Fundamentals



Amplifier Fundamentals

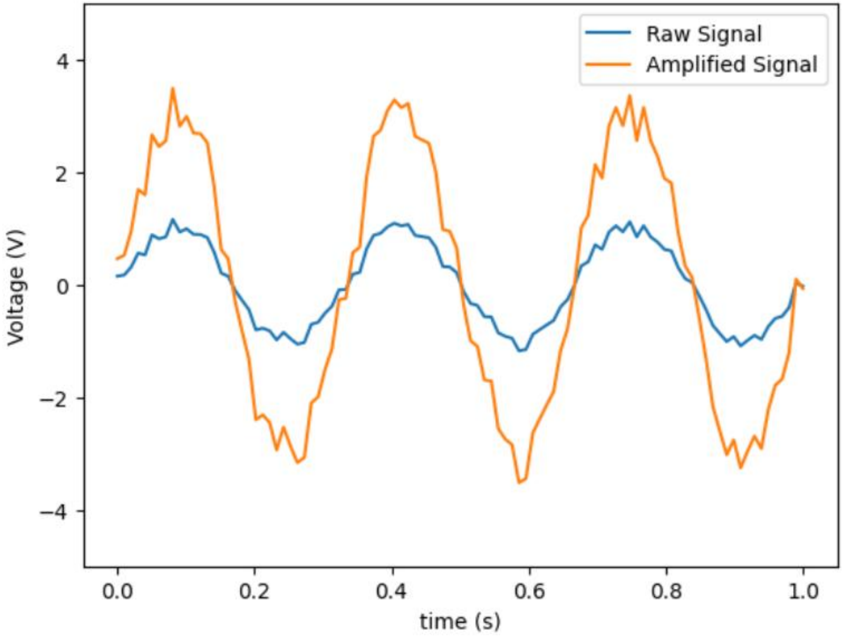
$$I = \frac{V}{R}$$

Constant Voltage → Changing Resistance → Changing Current that carries the signal from the Gate



Amplifier Fundamentals

- Gain: Ratio of the input and output signal amplitudes
- Linearity: Is the gain constant?
- Bandwidth: The signal frequency range that the amplifier has the best performance



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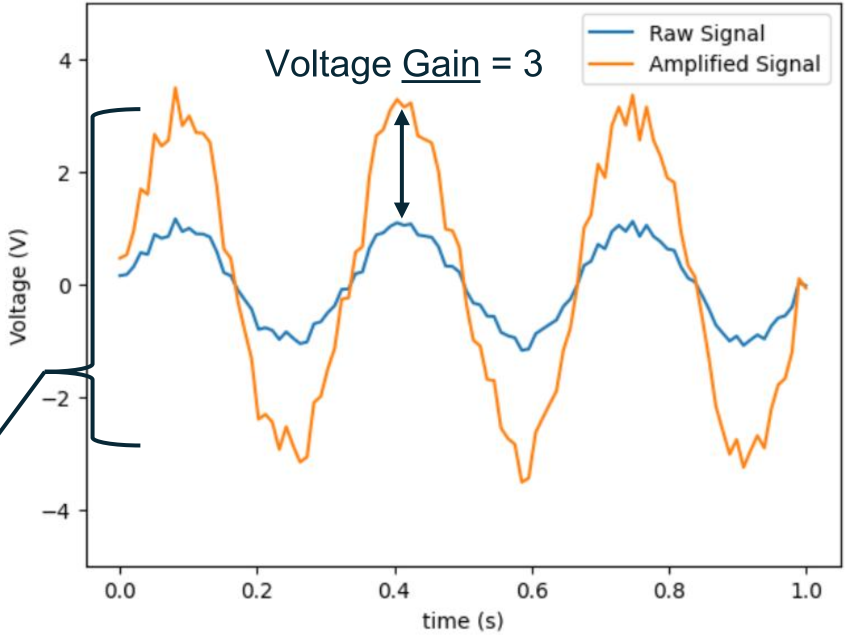


This amplifier is linear for these input voltages because each one becomes $V_{in} \rightarrow V_{out} = 3*V_{in}$

Gain

Power Gain and Voltage (Amplitude) Gain

$$\boxed{Power\ gain, Decibel = 10\log_{10}\left(\frac{P_{out}}{P_{in}}\right)}$$

$$Voltage\ gain, Decibel = 10\log_{10}\left(\frac{P_{out}}{P_{in}}\right)$$

$$= 10\log_{10}\left(\frac{\frac{V_{out}^2}{R}}{\frac{V_{in}^2}{R}}\right)$$

$$= 10\log_{10}\left(\frac{V_{out}^2}{V_{in}^2}\right)$$

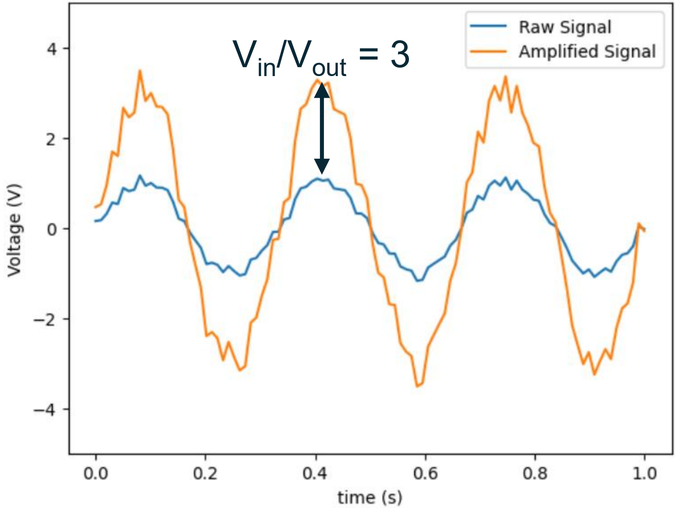
$$= 10\log_{10}\left(\frac{V_{out}}{V_{in}}\right)^2$$

$$\boxed{Voltage\ gain, dB = 20\ \log_{10}\left(\frac{V_{out}}{V_{in}}\right)}$$

Gain

$$Power\ gain, Decibel = 10\log_{10}\left(\frac{P_{out}}{P_{in}}\right)$$

$$Voltage\ gain, dB = 20\ \log_{10}\left(\frac{V_{out}}{V_{in}}\right)$$



The gain of the amplifier is:

$20\ \log_{10}(V_{in}/V_{out})$ dB = $20\ \log_{10}(3)$ dB

= 9.5 dB

Gain

- Power often uses units of **dBm** (decibel-milliwatts).
- We saw **dB** is a relative measurement between two powers

$$Power\ gain, Decibel = 10\log_{10}\left(\frac{P_{out}}{P_{in}}\right)$$

- If we set the reference power P_{in}= 1 mW, we get the decibel-milliwatts. So to convert to dBm,

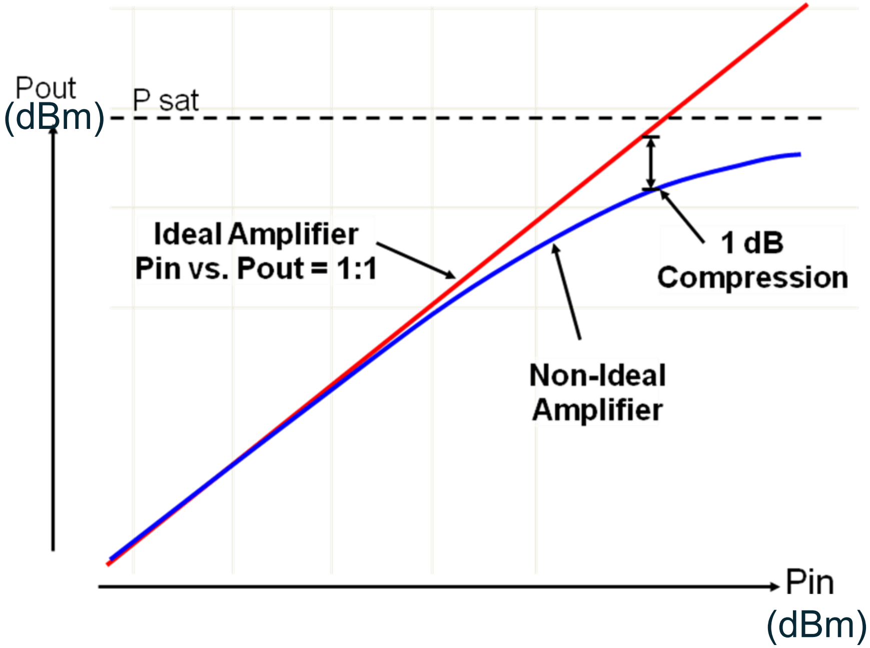
$$10\log_{10}\frac{P}{1\ \mathrm{mW}}$$

- Gain calculations are easy if you use these units!

  P_{in} = 0.1 mW = -10 dBm
  G = 20 dB
  → P_{out} = P_{in} + G = 10 dBm

Linearity

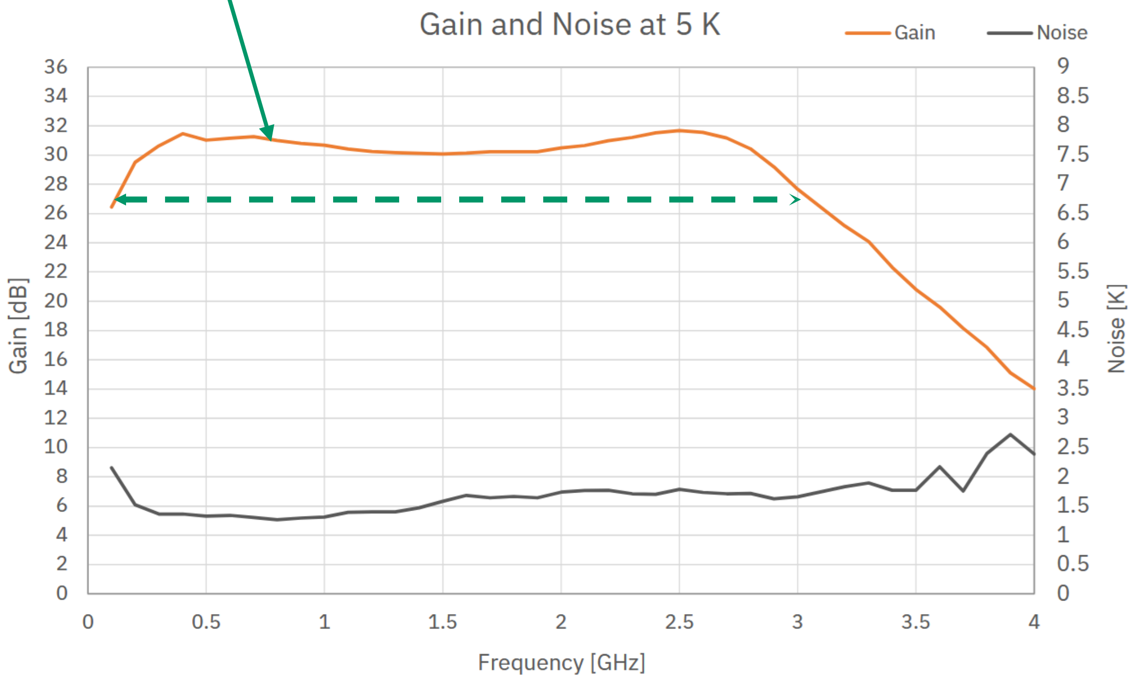
- For an ideal amplifier, you get the same gain for any signal you put in.
- In reality, the gain begins to decrease as the power increases.
- At the saturation power (P_{sat}), the amplifier can no longer provide gain



Bandwidth

- Bandwidth = Range where $G > G_0$-3dB (-3dB = Half Power)
- The advertised in-band gain is 30 dB, so the bandwidth is the frequency range where G > 27dB

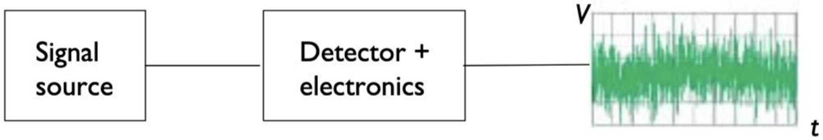
Product Features	
RF Bandwidth	0.2-3 GHz
Noise Temperature	1.6 K
Noise Figure	0.02 dB
Gain	30 dB
DC power (typical)	V_{ds} = 1.1 V, I_{ds} = 23 mA*



Amplifier Noise

- Adding a power amplifier to your system will increase the noise in the system.
    - Fact of life set by quantum mechanics. But we can get very close to noiseless.
    - Any inefficiency in embedding the input RF signal into the high power provided to the amplifier will introduce noise.
- In RF Power (transistor) amplifiers:
    - Thermal fluctuations in the gate material → Colder = Better
    - Shot noise as electrons cross the depleted region

Signal vs Noise



For any sensor we're detecting, we want to:

- **amplify our signal** as much as possible
- **introduce as little noise** as possible

We need a quantity that characterizes our signal strength over our noise.

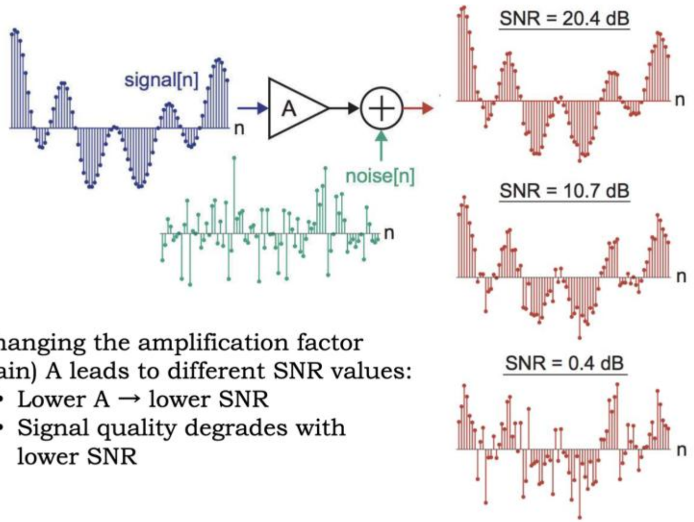
We will call it **Signal to Noise Ratio** (**SNR**).

Signal to Noise Ratio

- dB are useful for describing the ratio of different quantities. Instead of comparing input power to output power, we define **SNR** as the **ratio of signal power to noise power**.

$$\mathrm{SNR} = \frac{P_{\text{signal}}}{P_{\text{noise}}} = \left(\frac{A_{\text{signal}}}{A_{\text{noise}}}\right)^2$$

$$\mathrm{SNR}_{\mathrm{dB}} = P_{\text{signal,dB}} - P_{\text{noise,dB}}$$



Changing the amplification factor (gain) A leads to different SNR values:

- Lower A → lower SNR
- Signal quality degrades with lower SNR

Noise Factor and Noise Figure

Noise Factor of a device is:

$$\boxed{F = \frac{SNR_{in}}{SNR_{out}}}$$

Given that we know that all devices will add noise, what is the lowest possible noise factor we can have?

Noise Factor and Noise Figure

Noise Factor of a device is:

$$\boxed{F = \frac{SNR_{in}}{SNR_{out}}}$$

Given that we know that all devices will add noise, what is the lowest possible noise factor we can have?

SNR_out < SNR_in, so **F will always be >1**. We always want as low a noise factor as possible.

Noise Figure is the noise factor in decibels.

$$NF = 10 \log 10 \frac{SNR_{in}}{SNR_{out}} = SNR_{in,\mathrm{dB}} - SNR_{out,\mathrm{dB}}$$

Noise Figure and Noise Temperature

- We can think of **Noise Figure** in terms of the added noise from an amplifier.

$$NF = 10 \log_{10}(P_{noise\text{-}out}/P_{noise\text{-}in})$$

$$NF = 10 \log_{10}\{(\langle V^2_{noise}\rangle + \langle V^2_{amp}\rangle)/\langle V^2_{noise}\rangle\}$$

$$NF = 10 \log_{10}\{1 + \langle V^2_{amp}\rangle/\langle V^2_{noise}\rangle\}$$

- Noise is often measured in noise temperature, which equates add noise to the equivalent power of an ideal resistor

$$\frac{P_\mathrm{N}}{B} = k_\mathrm{B} T$$

- Thus,

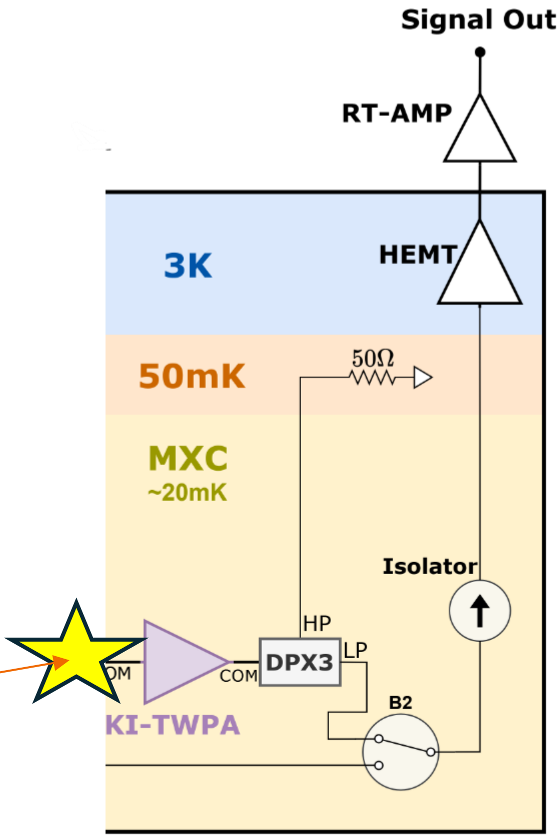
$$\mathrm{NF} = 10\log_{10}\left(1 + \frac{T_\mathrm{amp}}{T_\mathrm{source}}\right)$$

Amplifier Noise Cascade

- Detector readout systems may have many amplifiers
  - Can be used to reduce noise
  - Necessary from very weak signals

$$T_{\text{eq}} = T_1 + \frac{T_2}{G_1} + \frac{T_3}{G_1 G_2} + \cdots$$

Device (origin of signal)

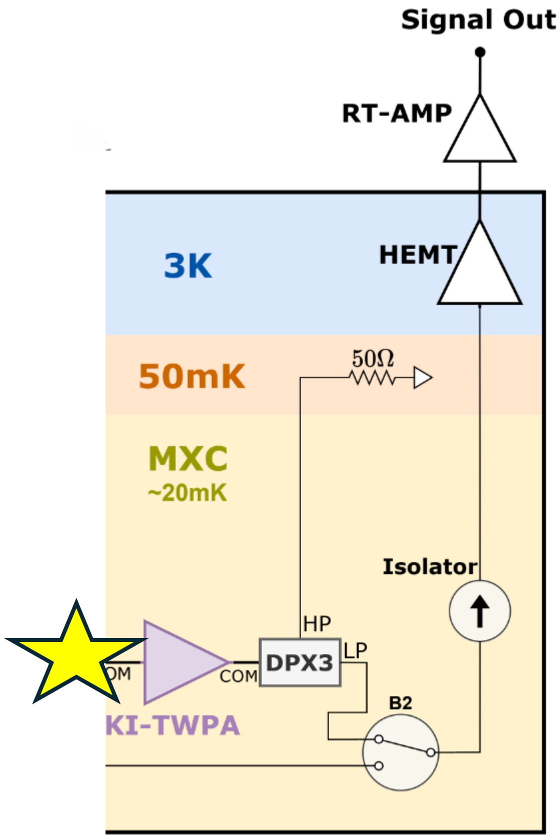


Amplifier Noise Cascade

- Amplifier noise is dominated by the first amplifier in the chain.

$$T_{\text{eq}} = T_1 + \frac{T_2}{G_1} + \frac{T_3}{G_1 G_2} + \cdots$$

- Suppose all 3 amps each have G = 30 dB.
  - T_1 ~ 50 mK (1 noise quanta at 1 GHz)
  - T_2 ~ 1.6 K
  - T_3 ~ 300 K
  - T_{eq}~ 0.5 K
- If you tried this without cryogenic amplifiers, T_1=T_2=T_3=300K and T_{eq}~300 K



Quantum Noise Limited Amplifiers

- Even if you had the perfect amplifier, it would introduce noise because of the Heisenberg Uncertainty Principle.
  - Cannot measure phase and amplitude at the same time without uncertainty.
- For a signal of frequency *f*, it has energy *hf* and a fundamental added noise of

$$T_{\mathrm{amp}} = \frac{hf}{2k_B}$$

- There is the same amount of noise for shot noise at the input.
- Total noise at the output of the amplifier is called a “noise quanta” – the smallest unit of noise for a given frequency

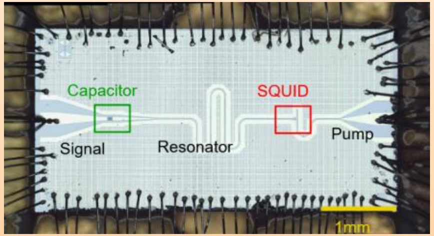
$$T_{\mathrm{total}} = T_{\mathrm{amp}} + T_{\mathrm{input}} = \frac{hf}{k_B}$$

Quantum Noise Limited Amplifiers

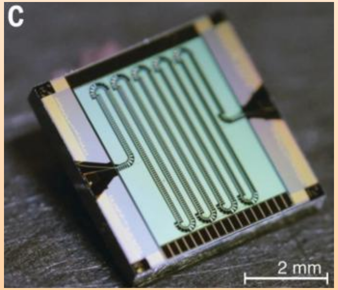
Nearly Quantum Limited

HEMT: 10s of noise quanta

Quantum Limited

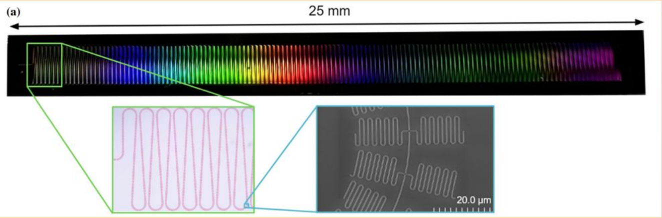


Josephson Parametric Amplifier



Josephson TWPA

Kinetic Inductance TWPA



Parametric Amplification

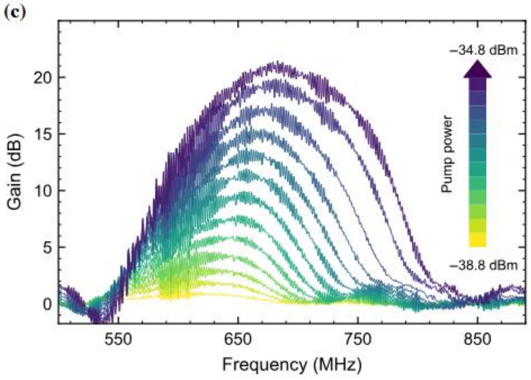
If you have ever played on a swing before, then you've already used a **parametric amplifier**!

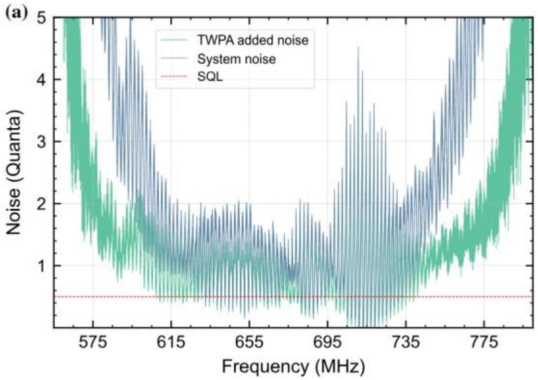
Parametric Amplification

- When you swing, your “signal” is the swing frequency and amplitude (height).
- You move your legs and body to increase the amplitude. This is the **pump signal**. By changing a parameter of the system (your center of mass) at the right frequency, you can swing higher and higher.

Parametric Amplifiers

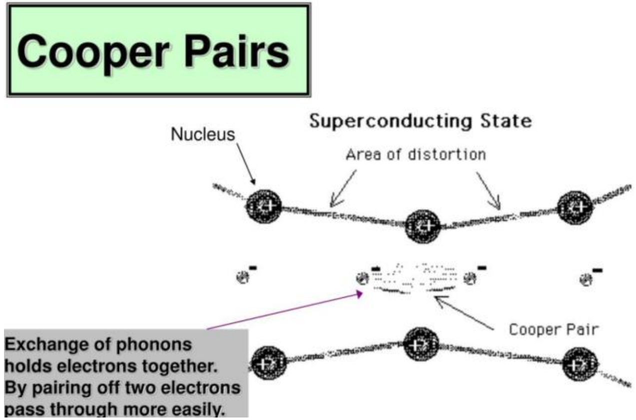
- Instead of changing the center of mass of the child on the swing, parametric amplifiers change the electrical reactance of a circuit.
- Because they are superconducting, these parametric amplifiers have almost no thermal noise. The modulation of the circuit’s reactance by the pump signal is extremely efficient, thus the amplifier can reach the quantum limit while providing 10s of dB of gain.





KI TWPAs

- Kinetic Inductance Traveling Wave Parametric Amplifiers
- The inertia of **cooper pairs** acts as an inductance called kinetic inductance. In narrow strips of thin superconducting films, this is the dominant inductance.
- **Non-linear** current dependence:
- A changing bias current causes amplification:



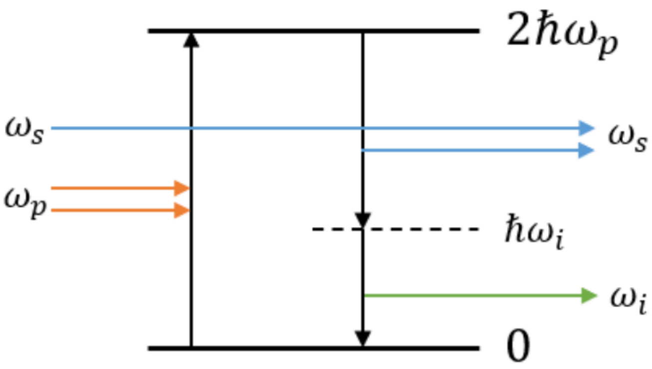
$$\mathcal{L}(I) \approx \mathcal{L}_0 \left(1 + \left(\frac{I}{I_*}\right)^2 + \left(\frac{I}{I'_*}\right)^4 + \ldots\right)$$

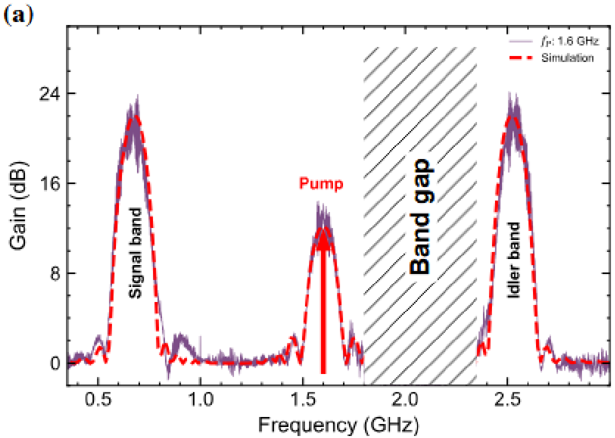
$$\begin{aligned} V &= \frac{d(LI)}{dt} \\ &= \frac{d}{dt}\left[L_0\left(1 + \left(\frac{I}{I_*}\right)^2 + \left(\frac{I}{I'_*}\right)^4\right) I\right] \\ &= \frac{d}{dt}\left[L_0 I + \alpha I^3 + \beta I^5\right] \end{aligned}$$

KI TWPAs

- **4 wave mixing**
- 2 photons from the pump are annihilated. By energy conservation, a signal band photon and an idler band photon are created.

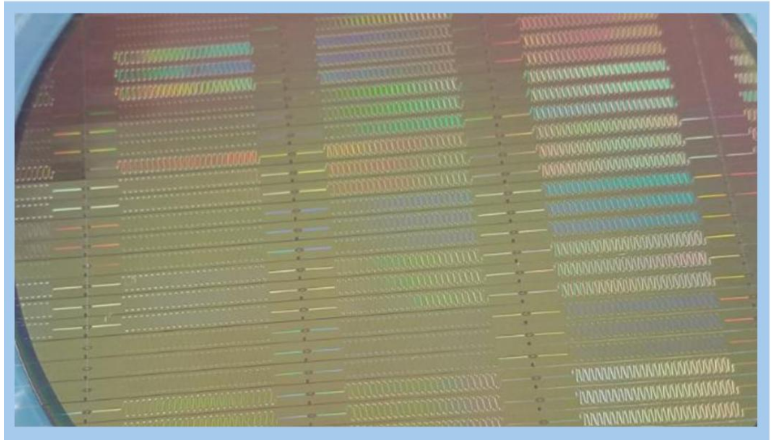
$$2\omega_p = \omega_s + \omega_i$$

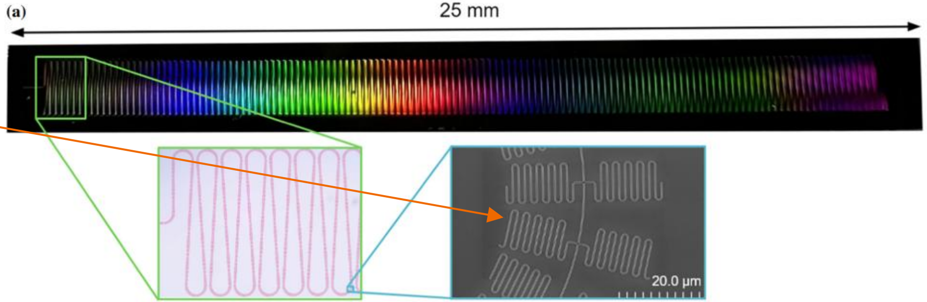




KI TWPAs

- KI TWPAs are long, meandering transmission lines.
- For a TWPA, waves need to travel the length of the device without high efficiency. The fingers you see here are dispersion engineered to phase match the signal along the entire transmission line.





KI TWPAs

- Broadband
- You can get **more gain** just by making the transmission line length longer.

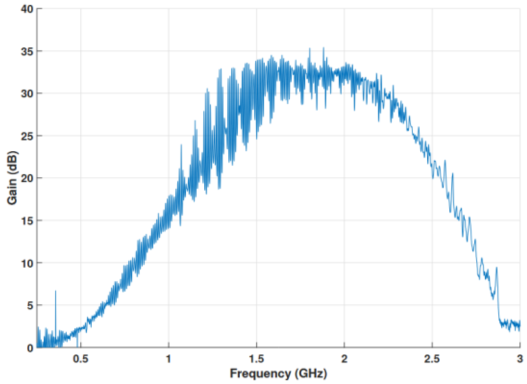
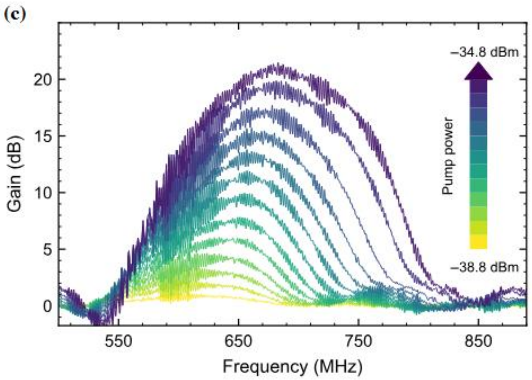
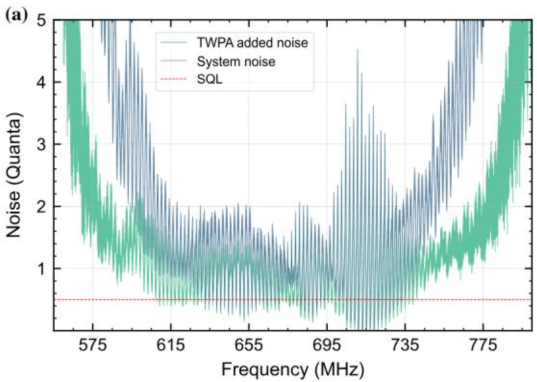


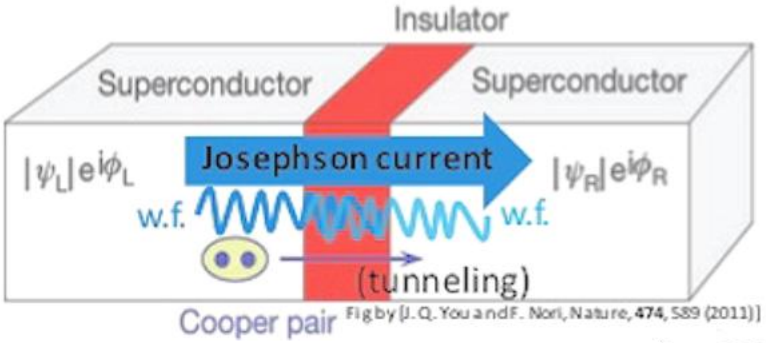
Figure 4.10: Four-wave-mixing gain from a 4 GHz pump measured at 4K.





Josephson Parametric Amplifiers

- **Josephson effect:** If two superconductors are separated by a very thin barrier, cooper pairs can tunnel across the barrier.



Josephson juntion

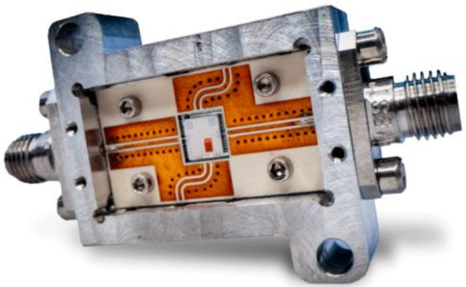
- This forms an inductor with a non-linear current with respect to phase Φ:

$$I = I_* \sin(\Phi)$$

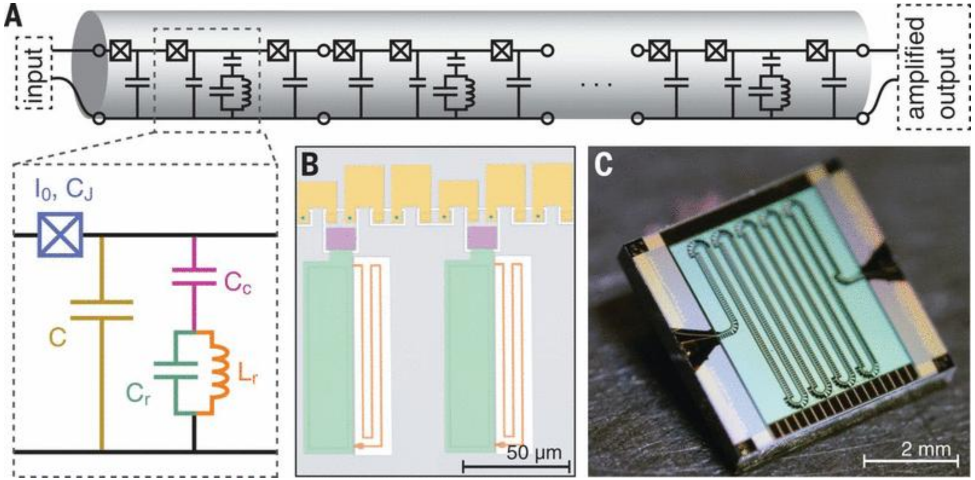
Josephson Parametric Amplifiers

- Josephson junctions act as resonators that amplify the signal when phase matched.
- Single resonator amp -> JPA
- Many junctions dispersed along a transmission line -> J TWPA

JPA



J-TWPA



Parametric Amplifiers

- JPA:
  - narrow band
  - compact
  - used for quantum computing applications with few qubits
- J-TWPA:
  - broadband
  - used for highly multiplexed qubit systems
- KI-TWPA:
  - ever more broadband
  - better magnetic field resilience
  - used for highly multiplexed qubit and detector systems
  - easier to fabricate